ECE 340 # Homework Assignment #1 **Fall 2026**

DUE: FRIDAY, SEPTEMBER 04, 2026

MAKE SURE YOUR **NAME** AND **NETID** ARE CLEARLY VISIBLE. SUBMIT ASSIGNMENTS AS A PDF (EITHER SCANNED OR TYPED). SHOW ALL WORK (PARTIAL CREDIT CANNOT BE GIVEN WITHOUT WORK). SHOW ALL UNITS. HOMEWORK MUST BE TURNED IN THROUGH GRADESCOPE. HOMEWORK SHOULD BE TURNED IN ON TIME AT 5:00PM ON THE DUE DATE. PLEASE CONTACT THE COURSE DIRECTOR, PROFESSOR DALLESASSE, SHOULD ANY ISSUES WITH LATE HOMEWORK ARISE.

1. UNITS AND MEASUREMENTS IN SEMICONDUCTORS

(A). In this course, we will consider often the concept of energy of a photon. A convenient formula for the photon energy in vacuum (E_{ph}), with the knowledge of the wavelength (λ), is:

$$E_{\text{ph}} = \frac{1.24\,(\text{eV}\cdot\mu\text{m})}{\lambda(\mu\text{m})} = \frac{1240\,(\text{eV}\cdot\text{nm})}{\lambda(\text{nm})} \tag{1.1}$$

Derive this formula, knowing that

$$\lambda = \frac{c}{\nu} \tag{1.2}$$

where c is the speed of light, ν is the photon frequency, and

$$E_{\text{ph}} = h\nu \tag{1.3}$$

where h is Planck's constant.

(B). Planck's constant can be expressed either in terms of Joules-second (J·s) or of electronvolts-second (eV·s). Reflect on which of the two versions of Planck's constant will be more convenient for your calculations. You can find Planck's constant values in Appendix II of the textbook, in Wikipedia entries, and many other sources. Take the ratio between the values of Planck's constant (in J·s) and the one in (eV·s). What value do you get and what does it represent?

(C). Consider a photon with energy 1.20 eV. What is the wavelength (in nm) of that photon?

(D). Measurements of semiconductor properties are often conducted at reference temperatures that can be easily reproduced at atmospheric pressure (note, you should always use Kelvin units for temperature). Many formulas used in semiconductor theory refer to the so-called "thermal voltage":

$$V_{\text{th}}(\text{eV}) = k_B(\text{eV/K}) \times T(\text{K}) = k_B T \tag{1.4}$$

where k_B is Boltzmann's constant and T is the temperature. Calculate the value of the thermal voltage at the following temperatures:

1. Standard room temperature (300 K).
2. 120 K.
3. Liquid nitrogen temperature (find this temperature value online).
4. Liquid helium temperature (find this temperature value online).

Solution **20 points in total**

(A). Rearrange the first formula for ν:

$$\nu = \frac{c}{\lambda} \tag{1.5}$$

and plug it into the second equation:

$$E_{\text{ph}} = \frac{hc}{\lambda} = \frac{(4.136\times10^{-15}\,\text{eV}\cdot\text{s})(2.998\times10^{8}\,\text{m/s})}{\lambda} = \frac{1240\,\text{eV}\cdot\text{nm}}{\lambda}. \tag{1.6}$$

5 points in total

(B). We will be working with electronvolts (eV) since we will often consider the energies of individual electrons. The electronvolt by definition is the amount of potential energy of a single electron under the electrostatic potential of one volt. Therefore, the energy of an electronvolt is 1 V times the charge of a single electron ($q = 1.602\times10^{-19}$ C). Since the definition of 1 V is 1 J/C (joules per coulomb), it follows that the ratio of J/eV is simply the magnitude of q, the elementary charge. This will be a useful constant to remember.
5 points in total

(C).

$$\lambda = \frac{1240\,\text{eV}\cdot\text{nm}}{1.2\,\text{eV}} = 1033\,\text{nm}. \tag{1.7}$$

2 points in total

(D). **8 points in total**

1. Room temperature (300 K):

$$V_{\text{th}} = (8.617\times10^{-5}\,\text{eV/K})\times300\,\text{K} = 25.9\,\text{meV}. \quad \textbf{2 points}$$

2. 120 K:

$$V_{\text{th}} = (8.617\times10^{-5}\,\text{eV/K})\times120\,\text{K} = 10.34\,\text{meV}. \quad \textbf{2 points}$$

3. 77 K:

$$V_{\text{th}} = (8.617\times10^{-5}\,\text{eV/K})\times77\,\text{K} = 6.6\,\text{meV}. \quad \textbf{2 points}$$

4. 4 K:

$$V_{\text{th}} = (8.617\times10^{-5}\,\text{eV/K})\times4\,\text{K} = 0.345\,\text{meV}. \quad \textbf{2 points}$$

2. Lattice and Crystal Structures

Complete the following questions for 1) Simple Cubic (SC) Lattice, 2) Body-Centered Cubic (BCC) Lattice, 3) Face-Centered Cubic (FCC) Lattice and 4) Diamond Crystal Structure.

(A). Calculate the total number of atoms found inside the cubic unit cell. Pay attention to the boundary of unit cell for each case.

(B). Calculate the nearest neighbor distance with the assumption of lattice constant of a.

(C). Which structure has the largest atomic density? Which one has the smallest atomic density? Specify the reason.

(D). For Simple Cubic Lattice, Body-Centered Cubic Lattice and Face-Centered Cubic Lattice, calculate the packing fraction (the maximum fractional volume of unit cell that are occupied by sphere (atoms)) of each lattice structure.

Solution

20 points in total

(A). **1 point each, 4 points in total**

(1) **SC lattice**

For a SC unit cell, each corner atom is only 1/8th contained in a unit cell (a corner atom is shared by 8 unit cells) , therefore the atom number is:

8 (corner atoms) × 1/8 = 1 atom in the SC unit cell.

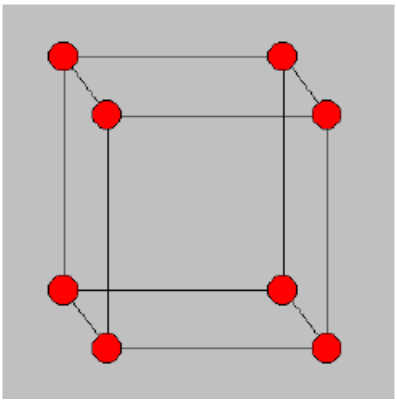


Figure 2.1: Simple cubic unit cell in real space.

(2) **BCC lattice**

Apart from the corner atoms contributing 1/8th of an atom, there is one atom in the center contributing a full atom to the unit cell. Therefore, the atom number is:

$$8 \times 1/8 + 1 \text{ (atom in the center)} = 2 \text{ atoms in BCC unit cell.}$$

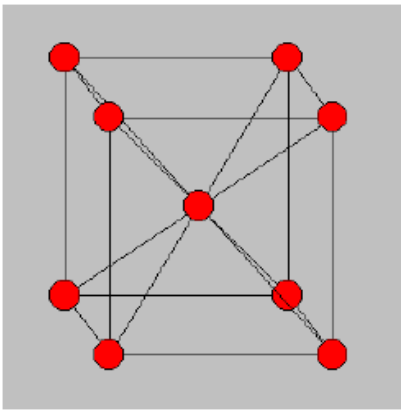


Figure 2.2: BCC unit cell in real space.

(3) **FCC lattice**

Apart from the eight corner atoms shared with neighboring unit cell, the atoms on each face are shared with two unit cells, therefore the atom number is:

$$1/8 \times 8 + 1/2 \times 6 = 4 \text{ atoms in the FCC unit cell.}$$

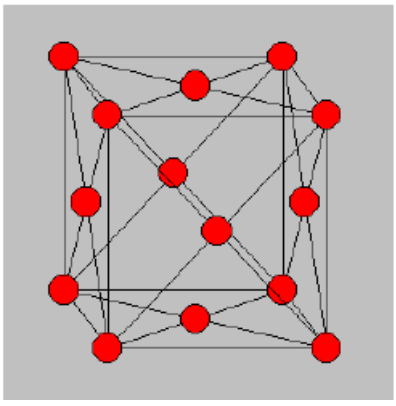


Figure 2.3: FCC unit cell in real space.

(4) **Diamond lattice**

A diamond crystal structure consists of two interpenetrating FCC lattices, separated by a distance in real space of a/4, a/4, a/4. From previous results, we know one FCC lattice contributing 4 atoms to the unit cell. Since the second FCC lattice has been shifted such that there are 4 whole atoms within a single unit cell that are not shared, so the atom number is:

$$1/8 \times 8 + 1/2 \times 6 + 4 = 8 \text{ atoms in the diamond lattice unit cell.}$$

(B). **1 point each, 4 points in total**

(1) **SC lattice**

It is apparent that the nearest neighbor to any atom in the unit cell is the atom at an adjacent corner along the edge of the unit cell which is the lattice constant *a*.

(2) **BCC lattice**

From Figure 2.2 we can see that the atom at the bottom left corner has a nearest neighboring atom which is at the center of the unit cell. The distance is:

$$\sqrt{(\frac{a}{2})^2 + (\frac{a}{2})^2 + (\frac{a}{2})^2} = \frac{a\sqrt{3}}{2}$$

(3) **FCC lattice**

From Figure 2.3, we can see that for any atoms at the corner of the unit cell, the nearest neighboring atom is the atom on the surface and therefore, the nearest distance is:

$$\sqrt{(\frac{a}{2})^2 + (\frac{a}{2})^2} = \frac{a\sqrt{2}}{2}$$

(4) **Diamond lattice**

The nearest neighbor from the bottom left corner atom is a/4 away in all directions. To visualize this, remember that the diamond structure is composed of two interpenetrating FCC lattices offset by a/4 in all directions. Therefore, the nearest neighbor distance is:

$$\sqrt{(\frac{a}{4})^2 + (\frac{a}{4})^2 + (\frac{a}{4})^2} = \frac{a\sqrt{3}}{4}$$

(C). **6 points total** The atom density is number of atoms divided by the volume of the unit cell. Given the lattice constant of *a*, the volume of the unit cell is a^3. Therefore, the structure with the highest number of atoms per unit cell will have the highest atomic density. Therefore the diamond lattice has the highest atomic density and the simple cubic has the lowest atomic density.

2 pts each for identified correct unit cell for highest and smallest atomic density, 2 pts for correct explanation

(D). **6 points in total** For the packing fraction, we assume that all atoms in the unit cell are hard, non-overlapping spheres. To be non-overlapping yet maximize the total volume enclosed in each unit cell, the radius of each sphere must be exactly half of the nearest neighbor distance you calculated in part (b). It is also important to note that the total volume of the unit cell for each lattice and crystal structure is always a^3. The equation for the packing fraction is as follows:

$$\text{Packing fraction (PF)} = \frac{\text{number of atoms per unit cell} \cdot \text{volume of sphere}}{\text{volume of unit cell}} \tag{2.1}$$

1. SC lattice packing fraction: **2 points**

$$\text{sphere radius} = \frac{a}{2} \tag{2.2}$$

$$\text{volume of sphere} = \frac{4}{3}\pi\left(\frac{a}{2}\right)^3 \tag{2.3}$$

$$\text{atoms per unit cell} = 1 \tag{2.4}$$

$$\boxed{\text{PF} = \frac{\pi}{6} \simeq 0.52 = 52\%} \tag{2.5}$$

2. BCC lattice packing fraction: **2 points**

$$\text{sphere radius} = \frac{a\sqrt{3}}{4} \tag{2.6}$$

$$\text{volume of sphere} = \frac{4}{3}\pi\left(\frac{a\sqrt{3}}{4}\right)^3 \tag{2.7}$$

$$\text{atoms per unit cell} = 2 \tag{2.8}$$

$$\boxed{\text{PF} = \frac{\pi\sqrt{3}}{8} \simeq 0.68 = 68\%} \tag{2.9}$$

3. FCC lattice packing fraction: **2 points**

$$\text{sphere radius} = \frac{a\sqrt{2}}{4} \tag{2.10}$$

$$\text{volume of sphere} = \frac{4}{3}\pi\left(\frac{a\sqrt{2}}{4}\right)^3 \tag{2.11}$$

$$\text{atoms per unit cell} = 4 \tag{2.12}$$

$$\boxed{\text{PF} = \frac{\pi\sqrt{2}}{6} \simeq 0.74 = 74\%} \tag{2.13}$$

3. Density of Crystals and Bonding

(A). Calculate the (mass) densities $\left(\frac{\text{g}}{\text{cm}^3}\right)$ of Si and InP from their lattice constants (Appendix III), atomic weights, and Avogadro's number. Qualitatively compare the results with the densities given in Appendix III (no written comparison needed).

(B). For the following commonly used materials determine the bonding type (covalent, mixed ionic-covalent, metallic, and ionic). No explanation needed.

1. Si
2. InP
3. NaCl
4. Au
5. P

Solution **20 points in total**

(A). **Si:**

$$\text{atomic weight} = 28.1\,\text{amu} \tag{3.1}$$

$$\text{lattice constant of unit cell} = 5.43\,\text{Å} = 0.543\,\text{nm} = 5.43\times10^{-8}\,\text{cm} \tag{3.2}$$

$$\text{atoms per unit cell} = 8 \tag{3.3}$$

$$\text{mass density} = \frac{8\times28.1\times1.66\times10^{-24}\,\text{g}}{\left(5.43\times10^{-8}\,\text{cm}\right)^3} = 2.33\,\text{g/cm}^3. \tag{3.4}$$

5 pts total: 1 pts for atomic weight, 1 pts for lattice constant, 1 pts for atoms per unit cell, 2 pts for correct mass density.

InP:

$$\text{In atomic weight} = 114.82\,\text{amu} \tag{3.5}$$

$$\text{P atomic weight} = 30.97\,\text{amu} \tag{3.6}$$

$$\text{lattice constant of unit cell} = 5.87\,\text{Å} = 0.587\,\text{nm} = 5.87\times10^{-8}\,\text{cm} \tag{3.7}$$

$$\text{atoms per unit cell} = 8 \tag{3.8}$$

$$\text{mass density} = \frac{\left(4\times114.82+4\times30.97\right)\times1.66\times10^{-24}\,\text{g}}{\left(5.87\times10^{-8}\,\text{cm}\right)^3} = 4.80\,\text{g/cm}^3. \tag{3.9}$$

5 pts total: 1 pts for atomic weight, 1 pts for lattice constant, 1 pts for atoms per unit cell, 2 pts for correct mass density.

(B). **10 pts total:** 2 pts each for correct bonding of each materials.

1. **Si:** Covalent bonding
2. **InP:** Mixed ionic-covalent bonding
3. **NaCl:** Ionic Bonding
4. **Au:** Metallic Bonding
5. **P:** Covalent bonding

4. Epitaxial Growth

(A). Define the term Critical Thickness.

(B). Define the term Pseudomorphic Growth.

(C). Define the term Strained Layer Superlattice.

(D). Name one epitaxial growth technology.

Solution **20 points in total**

(A). Critical Thickness: Growth thickness above which misfit dislocations are formed. **5 pts**

(B). Pseudomorphic Growth: Epitaxial growth of a non lattice-matched material. **5 pts**

(C). Strained Layer Superlattice: A structure of alternating layers having tensile and compressive strain such that net strain balance occurs and misfit dislocations are not formed. **5 pts**

(D). For example: Molecular Beam Epitaxy (MBE), Metal-Organic Chemical Vapor Deposition (MOCVD), or Vapor Phase Epitaxy (VPE).
Bonus Material: Among these three examples, MOCVD and MBE are more commonly used epitaxial growth techniques in III-V semiconductors.
5 pts as long as student named a valid epitaxial growth technique. However, no points will be given if student name bulk growth technique, for example Czochralski method.